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Sequences & Series

1. An infinite geometric series has first term a and common ratio r .

The sum to infinity of this series is S .

The sum to infinity of the squares of the terms of this series is $2S$.

The sum to infinity of the cubes of the terms of this series is $\frac{36}{7}S$.

What is the value of S ?

- A 2
- B 3
- C 4
- D 6
- E 8
- F 9
- G 12

Algebra

- 2.** A straight line with gradient m passes through the point $(1, 4)$

The line intersects the curve $y = x^2$ at two distinct points with x -coordinates x_1 and x_2

What is the minimum possible value of $x_1^2 + x_2^2$ as m varies over all real numbers?

- A** 1
- B** 4
- C** 7
- D** 8
- E** 12
- F** 16

Trigonometry

3. Let n be a positive integer.

Suppose that

$$\sum_{k=0}^{n-1} \cos\left(\theta + \frac{k\pi}{2}\right) = 0$$

for every real value of θ .

Which of the following is the complete set of possible values of n ?

- A** n is odd
- B** n is even
- C** n is a multiple of 3
- D** n is a multiple of 4
- E** n is 1 more than a multiple of 4
- F** n is 2 more than a multiple of 4

4. For real parameter t , consider the equation

$$\log_3 x + \log_x 3 = t$$

with $x > 0$ and $x \neq 1$.

The number of positive solutions x is

- A** two if $|t| > 2$, one if $|t| = 2$, and zero if $|t| < 2$
- B** one if $t > 0$, and zero if $t < 0$
- C** zero if $t > 2$, one if $t = 2$, and two if $0 < t < 2$
- D** one for all real t
- E** two if $t < -2$, zero if $t > -2$, and one if $t = -2$

Coordinate Geometry

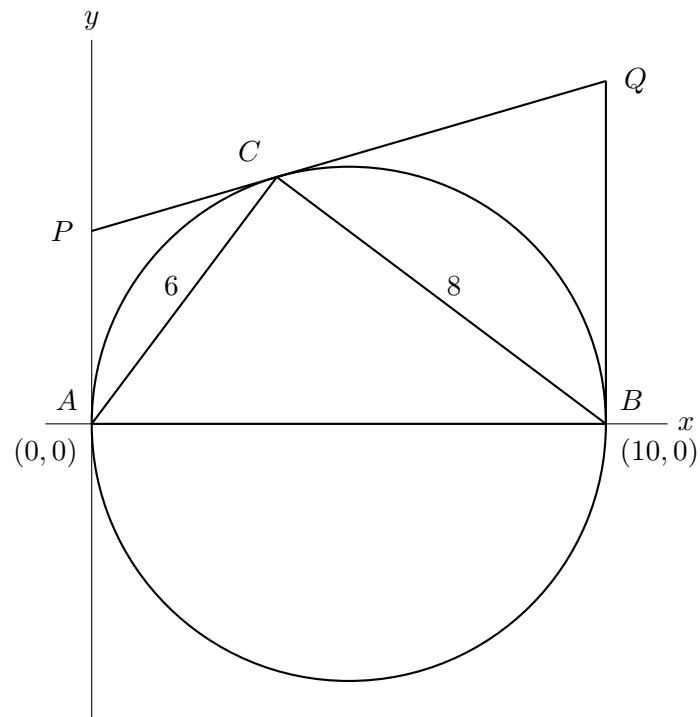
5. The points A and B have coordinates $(0,0)$ and $(10,0)$ respectively.

A circle has diameter AB .

The point C lies on the circle above the x -axis, with $AC = 6$ and $BC = 8$.

The tangent to the circle at C meets the lines $x = 0$ and $x = 10$ at P and Q respectively.

What is the length PQ ?



- A $\frac{35}{12}$
- B $\frac{15}{4}$
- C $\frac{20}{3}$
- D 10
- E $\frac{125}{12}$
- F $\frac{125}{6}$

Integration

6. Let k be a positive real constant. The function $f(x)$ is defined for $x > 0$ by

$$f(x) = \int_1^x \frac{t - k}{\sqrt{t}} \, dt$$

What is the complete set of values of k for which the equation $f(x) = 0$ has exactly two distinct real solutions for $x > 0$?

- A $k > 1$
- B $k > \frac{1}{3}$
- C $k > \frac{1}{3}$ and $k \neq 1$
- D $\frac{1}{3} < k < 1$
- E $k > 0$ and $k \neq 1$
- F $k \geq \frac{1}{4}$ and $k \neq 1$

Geometry

7. In triangle ABC , $AB = 9$, $\angle ABC = 30^\circ$, and $AC = x$, where $x > 0$.

For which values of x is there exactly one possible length of BC ?

A $x < \frac{9}{2}$

B $x = \frac{9}{2}$ or $x \geq 9$

C $\frac{9}{2} < x < 9$

D $x \geq \frac{9}{2}$

E $x = 9$ only

F $x > 0$

G $0 < x < \frac{9}{2}$ or $x > 9$

Probability & Counting

8. $N = 2^7 \times 3^5$

One of the positive factors of N that is divisible by 12 is selected at random, with each such factor equally likely to be selected.

What is the probability that N divided by the selected factor is a perfect square?

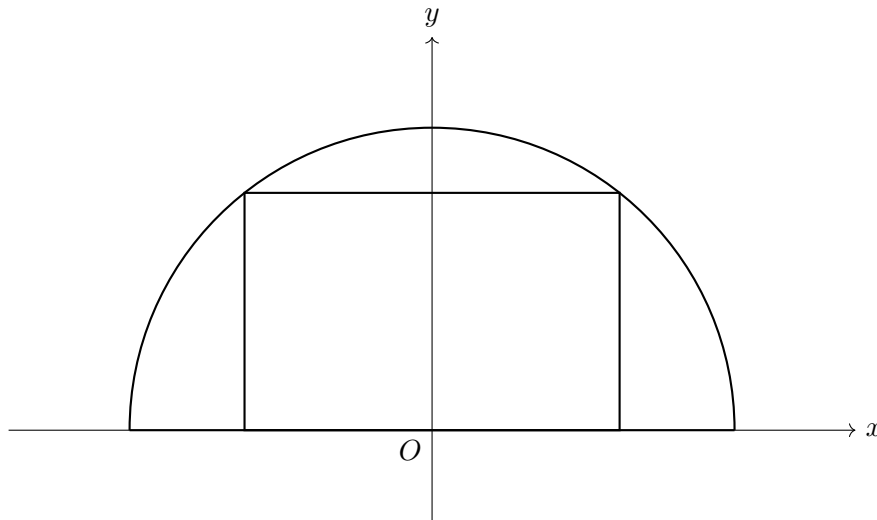
- A $\frac{1}{6}$
- B $\frac{1}{5}$
- C $\frac{1}{4}$
- D $\frac{3}{10}$
- E $\frac{1}{3}$
- F $\frac{2}{5}$

Geometry

9. The diagram shows a semicircle of radius 1 with its diameter along the x -axis and its centre at the origin.

A rectangle has one side on the diameter and its two remaining vertices on the arc of the semicircle.

What is the largest possible perimeter of such a rectangle?



- A $\sqrt{5}$
- B 4
- C $2 + \sqrt{5}$
- D $3\sqrt{2}$
- E $2\sqrt{5}$
- F 5

10. Consider the equation

$$\sin^2 x - (2k - 1) \sin x + k^2 - k = 0$$

where k is a real constant.

Find the complete set of values of k for which the equation has exactly three distinct real solutions in the range $0 \leq x < 2\pi$.

- A** $k = 0$ or $k = 1$
- B** $k = -1$ or $k = 2$
- C** $k = -1, k = 0, k = 1$, or $k = 2$
- D** $0 < k < 1$
- E** $-1 < k < 0$ or $1 < k < 2$
- F** $-1 < k < 1$
- G** $-1 \leq k \leq 1$
- H** There are no values of k .

11. A sequence a_n is defined by $a_1 = 1$ and, for $n \geq 1$,

$$a_{n+1} = a_n + \frac{1}{a_n}$$

Which of the following statements is true for all $n \geq 1$?

- A a_n is an integer
- B a_n^2 is an integer
- C $a_n < n$
- D $a_n^2 \geq 2n - 1$
- E $a_n^2 \leq 2n - 1$
- F $a_n \leq \sqrt{2n}$

Number

- 12.** The positive real numbers $p \times 10^{-4}$, $q \times 10^{-2}$ and $r \times 10^{-1}$ are each in standard form, and

$$(p \times 10^{-4}) + (q \times 10^{-2}) = (r \times 10^{-1})$$

How many of the following statements must be true?

I: $q > 1$

II: $r > 1$

III: $q < r$

IV: $p < r$

- A** 1 of them
B 2 of them
C 3 of them
D 4 of them
E None of them

Probability & Counting

- 13.** A bag contains only red balls and blue balls, with at least one ball of each colour. The total number of balls is n , where $30 < n < 50$.

Two balls are drawn at random without replacement. The probability that they have the same colour is equal to the probability that they have different colours.

How many different possible values are there for the number of red balls in the bag?

- A** 1
- B** 2
- C** 3
- D** 4
- E** 5

14. In this question, $a_1, \dots, a_{50}$ and $b_1, \dots, b_{50}$ are real numbers such that

$$a_n \leq 2b_n + 5$$

for each n . Consider the following statements:

- I:** (minimum of $a_1, \dots, a_{50}$) $\leq 2 \times$ (minimum of $b_1, \dots, b_{50}$) $+ 5$
- II:** (maximum of $a_1, \dots, a_{50}$) $\leq 2 \times$ (maximum of $b_1, \dots, b_{50}$) $+ 5$
- III:** (minimum of $a_1, \dots, a_{50}$) $\geq 2 \times$ (minimum of $b_1, \dots, b_{50}$) $+ 5$

Which of these statements must be true?

- A** None of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

Number

15. Let $N = 2^6 \times 3^4 \times 5^3$

A positive factor d of N is chosen such that the highest common factor of d and $\frac{N}{d}$ is 60.

How many possible values of d are there?

A 2

B 4

C 6

D 8

E 12

Polynomials

16. Consider

$$P(x) = x^4 + ax^3 + bx^2 + ax + 1$$

where a and b are real constants.

Suppose $(x - 2)$ and $(x + \frac{1}{2})$ are both factors of $P(x)$

What is the value of b ?

A -13

B -5

C 0

D 5

E 13

F $-\frac{17}{4}$

Integration

17. Let k be a positive constant. The curve C has equation

$$y = x^3 - 3k^2x + 2k^3$$

The point A is the local maximum of C .

The line parallel to the x -axis passing through A intersects C again at the point B .

The area of the region bounded by C and the line segment AB is 108.

What is the value of k ?

A 1

B $\sqrt{2}$

C $\sqrt{3}$

D 2

E $2\sqrt{2}$

F 3

G 4

Algebra

18. Let a and b be real numbers with $a + b = 1$

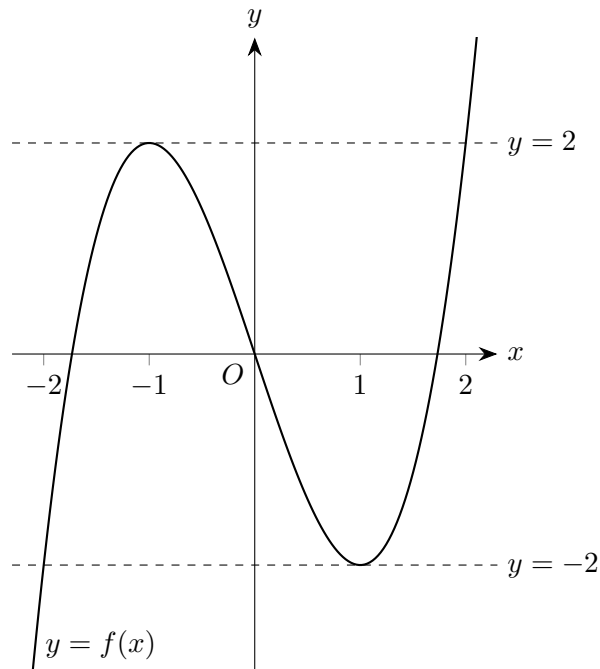
What is the least possible value of

$$(a^2 + b)(b^2 + a) \quad ?$$

- A $\frac{9}{16}$
- B $\frac{1}{4}$
- C $\frac{1}{2}$
- D $\frac{3}{4}$
- E 1

19. The diagram shows the graph of $y = f(x)$, where $f(x) = x^3 - 3x$. The horizontal lines through the turning points are also shown.

How many distinct real solutions does the equation $f(f(x)) = f(x)$ have?



- A 5
- B 6
- C 7
- D 8
- E 9

Inequalities

- 20.** Let $f(x) = ax^2 + bx + c$ be a quadratic function with real coefficients such that $f(x) \geq 0$ for all real x

Suppose that for every real x ,

$$f(x+1) - f(x) \geq 2x + 3$$

What is the smallest possible value of $f(0)$?

A 0

B $\frac{1}{4}$

C 1

D 2

E $\frac{9}{4}$

F 4

G $\frac{25}{4}$

Answer Key

Question	Answer
1	D
2	C
3	D
4	A
5	E
6	C
7	B
8	D
9	E
10	A
11	D
12	A
13	C
14	E
15	D
16	F
17	D
18	A
19	C
20	C